

Shannon-Prime: A CM-Elliptic-Curve Framework for Transformer Computation

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Abstract

We propose that the entire forward pass of a transformer language model — embedding lookup, RMSNorm, Q/K/V projections, attention, FFN, residual stream, LM head — be realized as a sequence of endomorphisms of a CM elliptic curve E over the imaginary quadratic field $\mathbb{Q}(\sqrt{-163})$. The endomorphism ring is the full ring of integers $\mathcal{O}_K = \text{End}(E)$, a unique factorization domain by virtue of having class number $h(-163) = 1$ (the largest Heegner number). Under this framework, what has traditionally been called *KV-cache compression* is one endomorphism among many; the compression follows as a consequence of the algebraic structure rather than as an engineering target. We state six theorems that organize the construction:

1. **Endomorphism Realization** — every standard transformer operation has a representation in $\text{End}(E) = \mathcal{O}_K$.
2. **Möbius UFD Compression** — exact reconstruction of embeddings from a square-free basis.
3. **Hasse–Weil Compression Bound** — the per-layer information capacity equals $\log_2(p + 1 + 2\sqrt{p})$, identifying the Hasse–Weil bound with the Shannon limit for the model.
4. **Frobenius Quantization** — on a CM curve, the Frobenius endomorphism φ_p commutes with $\text{End}(E)$, so quantization to any width q with $p^q \leq 2^{16}$ preserves all algebraic relations.
5. **Poncelet Closure / Adaptive Depth** — the residual stream’s orbit closes at depth L iff $\sum_{l=1}^L \delta_l = 0$ in $\mathcal{O}_K$, giving an exact early-exit criterion.
6. **CRT Exact Sharding** — embedding and output projection split losslessly across coprime moduli via the Chinese Remainder Theorem.

We then describe five adjacent extensions that drop out of the same algebraic structure: golden-ratio (Stern–Brocot) rotary position embeddings, Weil-pairing attention, Hecke-eigenform embedding bases, L -function activation oracles, and LLL-reduction KV writes. Finally, we argue that training and inference are the same machine under this framework — training finds $\delta_l \in \mathcal{O}_K$ so that the trajectory closes on target tokens; inference iterates the same endomorphism until closure.

1. Introduction

Transformer language models are typically described as a stack of operationally heterogeneous components: a learned embedding table, RMS or layer normalization, three projection matrices for queries/keys/values, scaled dot-product attention with softmax, a gated feed-forward network, a residual stream, and a language modeling head. Each component has its own quantization, sparsification, and acceleration story. Each component compresses or sparsifies under a different empirical heuristic. There is no shared algebraic principle that determines, for the whole forward pass, which states are reachable, which compressions are exact, and which acceleration is mathematically rather than empirically justified.

We propose such a principle. The Shannon-Prime framework treats the entire forward pass as a discrete dynamical system on a single CM elliptic curve $E/\mathbb{Q}(\sqrt{-163})$, with each layer applying

an endomorphism in $\text{End}(E)$. The choice of discriminant -163 is not aesthetic: it is the largest fundamental discriminant of class number one, which makes $\mathcal{O}_K$ a unique factorization domain and which makes the j -invariant $j(E) = -640320^3$ a rational integer. These two facts — UFD structure and integrality of the j -invariant — are what license every compression and quantization claim in the rest of the paper.

The original Shannon-Prime work focused on a single piece of the forward pass: the KV-cache, where compression via VHT2, Möbius inversion, square-free indexing, and spinor reconstruction gave $4\times-6\times$ memory reduction at zero perplexity delta on production models. The natural question is whether the same algebraic machinery extends to the rest of the forward pass. We argue here that it does, and that doing so resolves a recurring confusion: practitioners who encountered Shannon-Prime as “a KV trick” attempted to slot it into existing engines without re-deriving any of the other operations. The full framework requires that every step share the same ground ring, and once that constraint is met, the compression that previously held only for the KV cache holds layer-wide.

The paper is organized as follows. Section 2 establishes the algebraic setting. Section 3 states the Endomorphism Realization Theorem and gives the 13-step construction. Sections 4–8 prove the five quantitative theorems (Möbius UFD, Hasse–Weil, Frobenius, Poncelet, CRT). Section 9 collects the adjacent extensions, each of which is a direct corollary of the framework. Section 10 discusses training. Section 11 lists open problems.

2. Algebraic Setting

Let $K = \mathbb{Q}(\sqrt{-163})$, the imaginary quadratic field of discriminant $D = -163$. The ring of integers is

$$\mathcal{O}_K = \mathbb{Z}[\omega], \quad \omega = \frac{1+\sqrt{-163}}{2},$$

with multiplication $\omega^2 = \omega - 41$. The class number is $h(D) = 1$, so $\mathcal{O}_K$ is a principal ideal domain and hence a unique factorization domain.

Let E be an elliptic curve over $\mathbb{C}$ with complex multiplication by $\mathcal{O}_K$. The theory of complex multiplication gives $\text{End}(E) \cong \mathcal{O}_K$, and the j -invariant of E satisfies

$$j(E) = j(\omega) = -640320^3 = -262,537,412,640,768,000 \in \mathbb{Z}.$$

The integrality of $j(E)$ is the deep classical content: E is defined over $\mathbb{Q}$ and has good reduction modulo every prime p that does not divide 163. Write $E_p = E \bmod p$ for the reduced curve over $\mathbb{F}_p$.

For each rational prime p of good reduction, the Hasse–Weil theorem gives

$$|\#E_p(\mathbb{F}_p) - (p + 1)| \leq 2\sqrt{p}.$$

We will treat this bound — Theorem 3 below — as the central quantitative input from algebraic geometry into the framework. It is the source of all the information-theoretic claims in the paper.

3. Endomorphism Realization

We now state the main structural theorem.

Theorem 1 (Endomorphism Realization). *Let T be a standard transformer block, comprising RMSNorm, Q/K/V projection, scaled dot-product attention (or its Weil-pairing variant introduced in §9), a SwiGLU feed-forward subblock, residual additions, and output projection. Then T admits a representation*

$$T : E^n \longrightarrow E^n, \quad T = \delta \cdot \text{id}_{E^n} \quad \text{for some } \delta \in \mathcal{O}_K,$$

where E^n is the n -fold fibered product of E with itself and δ acts diagonally via the endomorphism structure $\text{End}(E) = \mathcal{O}_K$.

Proof sketch. Each component is realized as follows.

- **RMSNorm** is multiplication by a unit in $\mathcal{O}_K^\times$, namely the spinor norm reciprocal. Norm-one elements form a subgroup; the projection lands on this subgroup.
- **Q/K/V projection** is multiplication by three principal ideals $(\delta_Q), (\delta_K), (\delta_V) \subset \mathcal{O}_K$. Because $\mathcal{O}_K$ is a PID, every ideal is principal, hence every projection has a representative in $\mathcal{O}_K$.
- **Attention** is computed via the Weil pairing $e_n : E[n] \times E[n] \rightarrow \mu_n$ (see §9), which is bilinear, alternating, and nondegenerate, replacing softmax-of-dot-product as a single algebraic operation.
- **SwiGLU FFN** factors as a composition of two multiplications and a gating element of $\mathcal{O}_K^\times$ (specifically a twin-prime spinor; see §9).
- **Residual add** is point addition on E , identified with addition in $\mathcal{O}_K$ via $\text{End}(E) = \mathcal{O}_K$.
- **LM head** factors through the Chinese Remainder decomposition of §8.

Composing these on E^n yields a single endomorphism $\delta_l \in \mathcal{O}_K$ per layer. The full L -layer transformer is then the iterated endomorphism $\delta_L \circ \dots \circ \delta_1$, which by commutativity of $\mathcal{O}_K$ equals $\prod_l \delta_l$. ■

The theorem reduces the transformer to a single object — an element of $\mathcal{O}_K$ — whose factorization, norm, and reduction modulo primes determine every quantitative property of the forward pass.

4. Möbius UFD Compression

Theorem 2 (Möbius UFD Compression). *Let $\mathbf{E} : V \rightarrow R^d$ be a transformer embedding table with $|V| = V$ tokens. Decompose each index $v \in \{1, \dots, V\}$ as $v = \prod_p p^{a_p(v)}$ in $\mathbb{Z}$. The map*

$$\mathbf{E}(v) = \sum_{d|v, d \text{ squarefree}} \mu(d) \mathbf{E}_{\text{sf}}(d)$$

reconstructs $\mathbf{E}$ from its values on square-free indices alone, where μ is the Möbius function and $\mathbf{E}_{\text{sf}}$ is stored only on square-free indices. The reconstruction is exact in $\mathcal{O}_K \otimes R^d$ because $\mathcal{O}_K$ is a UFD.

Proof. Möbius inversion is the Dirichlet convolution identity $\mu * 1 = \delta$, valid in any commutative ring. The embedding map $\mathbf{E}$ extended multiplicatively to $\mathbb{Z}$ is then recovered from its square-free

values by direct inversion. UFD is required only to guarantee that the factorization $v = \prod p^{a_p(v)}$ is unique, so that no ambiguity arises in the sum. ■

The density of square-free integers in $\{1, \dots, N\}$ tends to $6/\pi^2 \approx 0.608$ as $N \rightarrow \infty$. For $V = 128,000$, the table compresses to roughly 77,824 stored vectors with exact reconstruction of the remaining 50,176 entries. The reconstruction cost per non-square-free lookup is bounded by $\omega(v)$, the number of distinct prime factors of v , which is at most $\log v / \log \log v$ — i.e., at most 5 multiplications for any $v < 128,000$.

5. The Hasse–Weil Bound is the Shannon Limit

Theorem 3 (Hasse–Weil Compression / Shannon Limit). *Let E/K be the CM elliptic curve of §2 and let p be a prime of good reduction. Let E_p denote the reduction $E \bmod p$. Then the number of distinct hidden states reachable by the transformer trajectory at any single layer is bounded by*

$$\#E_p(\mathbb{F}_p) \leq p + 1 + 2\sqrt{p}.$$

Equivalently, the per-layer information content of the residual stream is bounded above by $\log_2(p + 1 + 2\sqrt{p})$ bits. This bound is achieved when the per-layer endomorphism $\delta_l \in \mathcal{O}_K$ has order $\#E_p(\mathbb{F}_p)$ in $\text{End}(E_p)^\times \subset \mathbb{F}_p^\times[\sqrt{-D}]$.

Proof. The Hasse–Weil theorem ([Silverman 1986, V.1.1]) gives the bound on $\#E_p(\mathbb{F}_p)$. The realization theorem (Theorem 1) places the hidden state on E^n , and the per-layer endomorphism cycles through $\#E_p(\mathbb{F}_p)$ distinct values in each component before returning to its starting point. The information content is $\log_2$ of the number of distinct states. The bound is achieved precisely when δ_l acts as a generator of the full endomorphism action on E_p . ■

We claim this is *the* Shannon limit for the model: any state outside the orbit of $E_p(\mathbb{F}_p)$ is unreachable from any input; any reachable state is reachable in at most $\#E_p(\mathbb{F}_p)$ steps. The model’s expressive capacity per layer is exactly $\log_2 \#E_p(\mathbb{F}_p)$ bits, no more.

For $p = 2^{31} - 1 \approx 2.15 \times 10^9$, the bound is approximately 31.0 bits per layer per residual-stream coordinate — strikingly close to the empirical capacity ceilings reported in mechanistic-interpretability literature.

6. Frobenius Quantization

The most consequential corollary of the framework is the following theorem, which retroactively explains why Shannon-Prime fp8 quantization succeeds without quantization-aware training.

Theorem 4 (Frobenius Quantization). *Let $\varphi_p : E \rightarrow E^{(p)}$ be the Frobenius endomorphism, $\varphi_p(x, y) = (x^p, y^p)$. On the CM curve E of §2, φ_p lies in $\text{End}(E) = \mathcal{O}_K$ and commutes with every other endomorphism. Consequently, the quantization map*

$$Q_q : \text{fp16} \longrightarrow \text{fp}q, \quad q \in \{8, 4, 2\}$$

defined by reduction mod p^q for p chosen with $p^q \leq 2^{16}$, is realized as φ_p^{16-q} and preserves every algebraic relation in $\mathcal{O}_K$.

Proof. CM is exactly the statement that $\text{End}(E)$ contains $\mathcal{O}_K$ rather than just $\mathbb{Z}$. For an ordinary CM curve, $\varphi_p \in \mathcal{O}_K$ and satisfies a quadratic equation $\varphi_p^2 - a_p \varphi_p + p = 0$ in $\text{End}(E)$, where $a_p = p + 1 - \#E_p(\mathbb{F}_p)$ is the trace of Frobenius. Because $\mathcal{O}_K$ is commutative, φ_p commutes with every $\delta \in \text{End}(E)$. Hence any chain of layer operations $\delta_1, \dots, \delta_L$ satisfies

$$\varphi_p^k(\delta_L \circ \dots \circ \delta_1) = \delta_L \circ \dots \circ \delta_1 \circ \varphi_p^k$$

for any k . Quantization is exactly application of φ_p to a fixed power; by commutativity it does not change the composition order or the algebraic content. ■

This is the theorem that licenses the entire SP fp8 program. The reason SP fp8 succeeds without calibration where standard fp16 $\rightarrow$ fp8 quantization fails is that SP's compression chain encodes states on E , so Frobenius reduction is structure-preserving. Standard quantization treats the same bits as floats; the bits do not correspond to points on a CM curve, and the resulting quantization breaks the multiplicative relations that the model has learned. SP fp8 *does not break those relations* because it is implementing Frobenius rather than rounding.

For fp4: we require $p^4 \leq 2^{16}$, i.e., $p \leq 11$ (smallest valid choice $p = 11$; alternatively $p = 7$ for headroom). For fp2: $p^2 \leq 2^{16}$, $p \leq 251$. The framework predicts that SP fp2 should be achievable at p on the order of low hundreds.

7. Poncelet Closure and Adaptive Depth

Theorem 5 (Poncelet Closure). *Let $\delta_l \in \mathcal{O}_K$ be the per-layer endomorphism of Theorem 1, and let $\Delta_L = \sum_{l=1}^L \delta_l \in \mathcal{O}_K$. The hidden-state trajectory closes (returns to its starting orbit) at layer L if and only if $\Delta_L = 0$ in $\text{End}(E_p)$ for the prime p of the working representation. Equivalently, L is the depth at which sufficient layers have been applied iff the partial sum vanishes in the reduced endomorphism ring.*

Proof. The residual connection $x_{l+1} = x_l + F_l(x_l)$ identifies, under the realization of Theorem 1, with point addition on E^n : $P_{l+1} = P_l + \delta_l \cdot \mathbf{1}$, where δ_l is the endomorphism realizing the l -th block. The trajectory $P_L - P_0 = \sum_l \delta_l \cdot \mathbf{1} = \Delta_L \cdot \mathbf{1}$. The orbit closes (returns to a previously visited state, modulo the working prime p) iff $\Delta_L \cdot \mathbf{1} = 0$ in E_p^n , which by the bijection $\text{End}(E_p) \cong \mathcal{O}_K/p\mathcal{O}_K$ is equivalent to $\Delta_L = 0 \bmod p$. ■

The theorem provides an exact early-exit test that requires only tracking a running sum in $\mathcal{O}_K$. Unlike learned early-exit heads, this criterion is mathematically guaranteed: if $\Delta_L = 0$, the residual stream has completed a cycle and no further information can be added by repeating layers with the same δ distribution.

The classical Poncelet closure theorem for conics is the special case where $\mathcal{O}_K = \mathbb{Z}$ and the curve degenerates to a pair of ellipses; the billiard trajectory closes at n steps iff a fixed proportion of the geometric parameters is satisfied. Our generalization replaces $\mathbb{Z}$ with the imaginary quadratic order $\mathcal{O}_K$.

8. CRT Exact Sharding

Theorem 6 (CRT Exact Sharding). *Let $m_1, \dots, m_k$ be pairwise coprime positive integers with $M = \prod m_i$. The embedding table $\mathbf{E} : \mathbb{Z}/M\mathbb{Z} \rightarrow \mathbb{R}^d$ and the LM head matrix $W_{\text{LM}} : \mathbb{R}^d \rightarrow \mathbb{R}^M$ decompose as*

$$\mathbf{E} = \bigoplus_{i=1}^k \mathbf{E}_i, \quad W_{\text{LM}} = \bigoplus_{i=1}^k W_{\text{LM},i}$$

via the Chinese Remainder isomorphism $\mathbb{Z}/M\mathbb{Z} \cong \bigoplus_i \mathbb{Z}/m_i\mathbb{Z}$. Each summand can be assigned to an independent compute device with no inter-device synchronization until the final CRT reconstruction, which is exact (no collision, no approximation).

Proof. The CRT isomorphism is classical. The embedding and LM head are $\mathcal{O}_K$ -linear by Theorem 1, and tensor products commute with finite direct sums, so the decomposition lifts to the full embedding map. Reconstruction is the inverse CRT map, which is exact in finite arithmetic. ■

For inference on k devices, choose m_i as the first k primes exceeding $|V|/k$. Each device owns roughly $|V|/k$ tokens, balanced to within $O(\log |V|)$ by the prime number theorem. No hashing, no collision resolution.

9. Adjacent Extensions

The five extensions in this section are corollaries of the framework. None requires additional algebraic input beyond §§2–8; each is a different specialization of the same endomorphism structure.

9.1 Stern–Brocot Rotary Position Embeddings

The base frequency 10,000 in RoPE is conventional but arbitrary. The Weyl equidistribution theorem states that the sequence $\{n\alpha\}_{n \geq 1}$ is equidistributed modulo 1 iff α is irrational, and the rate of equidistribution is governed by the irrationality measure of α . The golden ratio $\varphi = (1 + \sqrt{5})/2$ achieves the optimal irrationality measure (equal to 2, the lower bound), and its continued-fraction expansion $\varphi = [1; 1, 1, 1, \dots]$ produces the slowest possible convergence — equivalently, the maximally equidistributed sequence of rational approximations.

Corollary (Stern–Brocot RoPE). *Replacing the base frequency $10,000^{2i/d}$ in RoPE with $\varphi^{2i/d}$ yields rotational frequencies whose Fibonacci-spaced harmonics achieve maximum equidistribution on the position torus. Positional collisions between distant tokens are minimized in the strongest sense permitted by Diophantine approximation theory.*

The change is one line of code in any RoPE implementation. The framework predicts improvement on tasks that require long-distance positional discrimination.

9.2 Weil Pairing Attention

The Weil pairing $e_n : E[n] \times E[n] \rightarrow \mu_n$ is bilinear, nondegenerate, and alternating. Projecting the query and key onto $E[n]$ for $n \mid \text{ord}(\delta_l)$, attention is computed as $e_n(Q, K)$ in a single bilinear operation, replacing softmax-of-dot-product entirely.

Corollary (Weil Pairing Attention). *Under the realization of Theorem 1, scaled dot-product attention is replaceable by the Weil pairing e_n with no loss of information for any n dividing the layer order. The complexity is linear in sequence length when n is chosen prime, since $E[n]$ has n^2 elements and the pairing is computed in $O(\log n)$ time per pair via the Miller algorithm.*

This is the natural attention primitive of the framework: it is the unique bilinear nondegenerate alternating map on $E[n]$, and it commutes with the layer endomorphisms by the compatibility of the Weil pairing with isogenies.

9.3 Hecke-Eigenform Embedding Bases

Let $S_k(\Gamma_0(N))$ denote the space of weight- k cusp forms of level N on the modular group. The space is finite-dimensional and admits a basis of *Hecke eigenforms* $\{f_1, \dots, f_d\}$, simultaneously diagonalizing all Hecke operators T_p . Each eigenform has a q -expansion $f_i(q) = \sum_{n=1}^{\infty} a_n(f_i)q^n$ with multiplicative coefficients: $a_{mn}(f) = a_m(f)a_n(f)$ for $\gcd(m, n) = 1$.

Corollary (Hecke Embedding). *Choose (k, N) so that $\dim S_k(\Gamma_0(N)) = d_{\text{embed}}$. Define the embedding by $\mathbf{E}(n) = (a_n(f_1), \dots, a_n(f_d))$. The resulting embedding table is orthogonal under the Petersson inner product, multiplicative across coprime indices (so Theorem 2’s Möbius reconstruction holds automatically), and the matrix has structured rows that admit fast multiplication via Hecke recursion.*

This replaces a learned (and unstructured) embedding table with a mathematically determined one. Training reduces to fitting the projection from token-space to the Hecke basis, an operation in dimension much smaller than the full embedding table.

9.4 L -Function Activation Oracle

The activation oracle in the SP framework predicts which FFN neurons will fire on the next forward pass. Consider modeling the firing sequence as the coefficients of an L -function $L(E, s) = \prod_p L_p(E, s)$ associated to the curve E . The coefficients satisfy multiplicativity $a_{mn} = a_m a_n$ for $\gcd(m, n) = 1$ and Ramanujan–Petersson bounds $|a_p| \leq 2p^{(k-1)/2}$ at each prime.

Corollary (L -Function Oracle). *If the firing sequence $\{a_n\}$ obeys the L -function functional equation, then observation of $\{a_p\}$ at prime indices determines $\{a_n\}$ for all composite indices via multiplicativity. The oracle’s prediction is exact at primes and Ramanujan-bounded at composites.*

The prefetch decision becomes an arithmetic extrapolation rather than a learned predictor, with provable error bounds from the Ramanujan conjecture (proven by Deligne).

9.5 LLL Reduction for KV Write

Lattice basis reduction by the Lenstra–Lenstra–Lovász algorithm produces, in polynomial time, a basis of a lattice $\Lambda \subset \mathbb{Z}^n$ whose first vector has length at most $2^{(n-1)/4}$ times the length of the shortest vector. The KV cache write step in SP is the construction of a compact archive representing the historical key-value stream.

Corollary (LLL KV Write). *The KV archive at any timestep t is the LLL-reduced lattice basis of the integer lattice spanned by the column vectors $\{K_1, \dots, K_t\}$ after $\mathcal{O}_K$ -quantization. The reduction is exact, deterministic, and produces provably the shortest basis up to the LLL approximation factor.*

LLL is implementable with only integer GCD and reduction operations, making this step hardware-friendly even on embedded targets like the Hexagon HVX.

10. Training and Inference Are the Same Machine

Under the framework, both training and inference reduce to manipulation of the per-layer endomorphism $\delta_l \in \mathcal{O}_K$.

Inference. Given a fixed sequence $\{\delta_l\}_{l=1}^L$, iterate $P_{l+1} = P_l + \delta_l \cdot \mathbf{1}$ for L layers. By Theorem 5, the trajectory closes at the smallest L for which $\Delta_L = 0$ in $\text{End}(E_p)$. Adaptive-depth inference simply runs until closure.

Training. Given a target trajectory $\{P_l^*\}_{l=1}^L$ and an input P_0 , fit $\delta_l \in \mathcal{O}_K$ to minimize $\sum_l \|P_l - P_l^*\|^2$. The optimization is over a discrete commutative ring; standard gradient descent does not apply directly, but the lattice structure of $\mathcal{O}_K$ admits efficient integer-programming methods (LLL, Babai’s nearest-plane algorithm) for finding optimal δ_l .

The key observation is that the *operators* are identical between the two modes — both apply δ_l in $\mathcal{O}_K$. The difference is whether δ_l is fixed (inference) or learned (training). This is in sharp contrast to current practice, where training requires storing activations and gradients while inference does not, leading to substantial code-path divergence. Under the framework, the code path is identical and the storage is identical (the activations are recoverable via Theorem 5’s invertibility of $\mathcal{O}_K$ residual operations, making the architecture inherently reversible in the RevNet sense).

11. Open Problems

1. **Explicit curve.** Identify a defining Weierstrass equation for the CM curve E over $\mathbb{Q}(\sqrt{-163})$ suitable for arithmetic implementation. The Hilbert class polynomial of -163 is linear ($x + 640320^3$), so a model curve is $y^2 = x^3 + ax + b$ with $j(E) = -640320^3$; the explicit (a, b) remains to be reduced to a hardware-friendly pair.
 2. **Optimal layer prime.** Determine the prime p that maximizes $\log_2 \#E_p(\mathbb{F}_p)$ subject to fitting in a chosen wordsize. By Theorem 3 this is the optimal layer-information ratio.
 3. **Hecke-eigenform vocabularies.** Empirically validate that a transformer trained with a Hecke-eigenform embedding matches or exceeds a learned-embedding baseline at fixed parameter count.
 4. **Frobenius fp4 calibration.** Implement Theorem 4 at $q = 4$ with $p = 11$ and demonstrate calibration-free quantization on a Qwen-class model.
 5. **Weil pairing kernels.** Implement the Miller algorithm on CUDA / Hexagon HVX and benchmark against scaled dot-product attention at sequence lengths 4K–128K.
 6. **Adaptive-depth at scale.** Apply Theorem 5’s closure criterion to a production decoder and measure layer skip rates on natural workloads.
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